

**25WS66 Coursework 2:
F537796**

Abstract

Unmanned Combat Air Vehicles have become increasingly useful in modern warfare. There is however a lack of suitable UCAV for naval based operations in most militaries. This paper identifies the need for a carrier-based fighter style multirole UCAV, and sets out a multi-objective design optimisation problem with a focus on aerodynamic performance. It shows a traceable SysML model which informs all design decisions, and presents a set of optimal design variables which can then be implemented in more detailed design.

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1 Introduction

1.1 Background

In recent years, it has become clear that autonomy in warfare is becoming the norm. Recent conflicts have seen unmanned vehicles engage in conflicts in unprecedented numbers, with significant consequences. On land, autonomous vehicles traverse battlefields to rescue wounded without putting medics at risk. At sea, autonomous vessels clear mines whilst keeping sailors away from the minefield. In the air, autonomous vehicles engage in a wide range of tasks from surveillance to area denial. The unmanned combat air vehicle (UCAV) is a concept that has been widely researched and considered for use by militaries [1]. UCAV is a broad term that could encompass a variety of roles, from single-use autonomous missiles to strategic bombers. Recently, the unmanned fighter has received attention, however, with BAe Systems unveiling their new Collaborative Combat Aircraft [2], which is designed to provide “electronic warfare and precision strike capabilities against airborne and ground targets.” This aircraft is intended to operate alongside Royal Air Force assets in frontline operations. This precludes it from use at sea.

Naval aviation is an important facet of a nation’s armed forces, with the United States centering their naval power around aircraft carriers [3]. There is precedent for UCAVs in a carrier’s arsenal, for example the X-47B [4]. Northrop Grumman however withdrew its bid for the project.

1.2 Problem Statement

As such, there is a significant opportunity for a carrier-based UCAV that can fill multiple roles. Standing off the aircrew from hazardous situations would reduce the risk to them, and would protect them from some of the harms of military flying (e.g. extreme g forces). This UCAV would need to provide QRA (Quick Response Alert), electronic warfare (EW), close air support (CAS), and surveillance roles to meet the demands of a modern navy. Moreover, the UCAV would have to excel in the combat air patrol (CAP) role, where it not only can protect an area of airspace supporting other operations, but can be used to keep a constant patrol of the asset from which it is launched (such as an aircraft carrier). With BAe systems developing a UCAV to contract, targeting foreign markets is desirable. Of these, France and the United States would be good contenders, thanks to their advanced naval assets and lack of alternative [4].

1.3 System Narrative

The UCAV must be a conventional aircraft with one or two modern low-bypass turbofan engines, to remain comparable to other aircraft in the fleet. It must be of similar size and weight as contemporary aircraft, and must be capable of operating from both an airstrip or an aircraft carrier. With the target being overseas navies, adapting to their carriers is necessary. As such, being able to operate both from an airfield but also from a CATOBAR (Catapult Assisted Take Off But Arrested Recovery) system is essential. It must be possible of engaging in CAP, staying on station for extended periods of time and engaging hostile targets both within and beyond visual range. Additionally, it must be able to intercept hostile targets quickly at a variety of ranges and altitudes. When engaged in frontline activities, it must be capable of supporting ground forces with the direction of a ground controller, performing electronic warfare, and surveilling the combat space for other forces.

1.4 Aims and Objectives

This paper aims to provide a set of design variables that produce a UCAV that meet the above narrative, and fill the niche in international naval air power described.

It intends to do this by:

1. Modelling a suitable architecture using Model Based Systems Engineering
2. Performing a multi-variable optimisation analysis on a selection of key design variables.
3. Having traceability from customer need to solution, to support negotiation between the design team and the customer.

2 Technical Results

2.1 Stakeholder and Non-functional Requirements Definition

As described in section REF, the UCAV is needed to fulfil a number of roles for different stakeholders. The primary stakeholders are: the operator; friendly parties; neutral parties; tactical commanders (TACOM); and interdisciplinary users (e.g. ground controllers). Additionally, the customer as an organisation has needs that must be fulfilled. These needs have been elicited as capabilities - what a stakeholder would “walk up” to the system to use it for. They can also be seen as the reasoning behind the functional requirements. In the axiomatic design methodology [5], this corresponds to the *Customer domain*. The perspective of the operator and TACOM have been prioritised, with those of ground controllers, other parties, and the customer considered but to a lesser extent. The capabilities hierarchy has been modelled in SysML using Enterprise Architect (see figure 1).

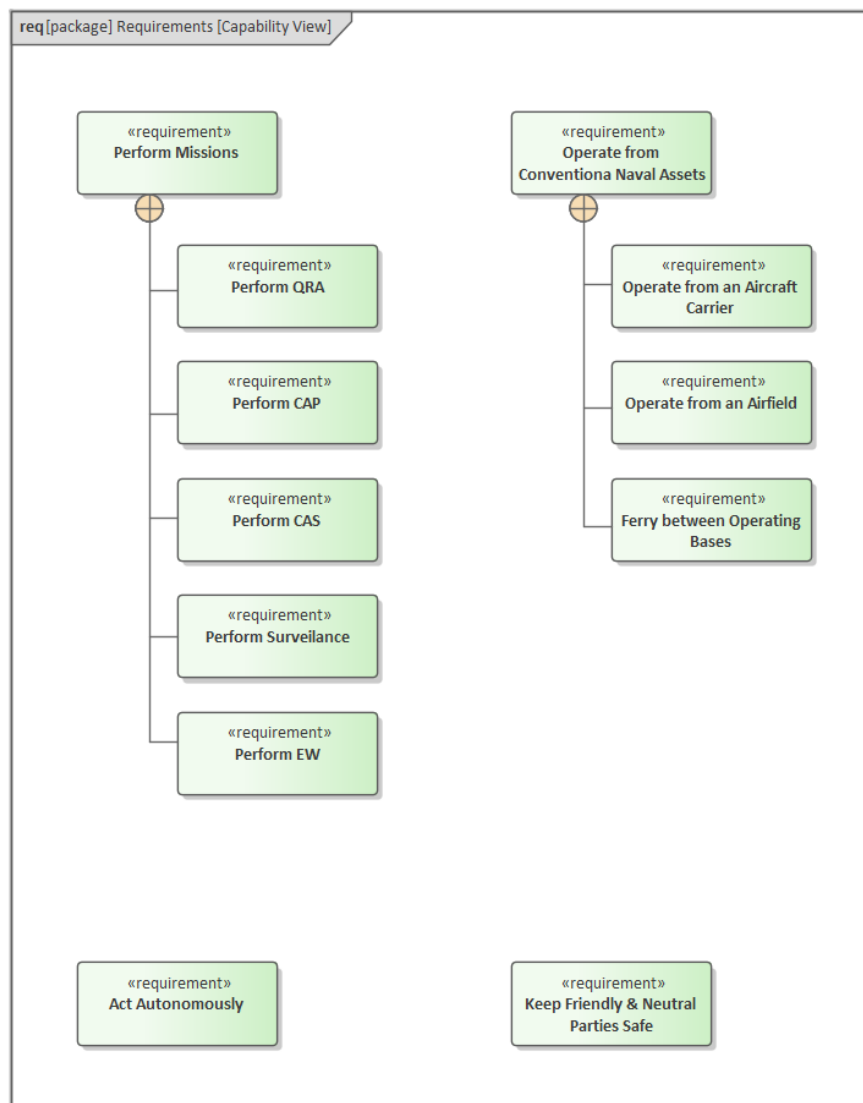


Figure 1: UCAV Capability View

This simple view highlights the main missions that the UCAV will need to undertake:

- Quick Reaction Alert (scrambling to interrogate a potential threat);
- Combat Air Patrol (loitering in an area to deter, detect, and defeat hostile targets);
- Close Air Support (striking ground targets near friendly land forces as directed by a controller);
- Surveillance (loitering or overflying an area to gather intelligence);

- Electronic Warfare (radiofrequency jamming).

It also captures some operational needs, key among which is operating from airfields or aircraft carriers. Of secondary importance at this stage of the design is autonomous control, which is intimately connected with the safety of friendly and neutral parties. As the design progresses and the wider system of systems is understood, these will likely into key capabilities that the UCAV must provide, and as such may be further decomposed. One shortfall of the capability model is the lack of reference to some significant capabilities. For example, cost would likely be a driving factor in the award of a contract by a customer so at this stage should be paramount. The model does however provide a good starting point for functional requirement elicitation and clear justification for them, from an aerodynamic design standpoint.

From these capabilities, a set of functional and non functional requirements have been derived. These are grouped into four on the diagram titled “Requirements”. The functional requirements under “mission effectiveness” were identified initially, and they are traceable back to the capabilities using «deriveReq» relationships. These relationships are hidden to improve readability of the diagram. As is shown in figure 2, a number of “Aerial Performance” requirements were then derived from these functions, with “Operation and Support Functions” and some other functional requirements in “Aerial Performance” being again derived from the capabilities.

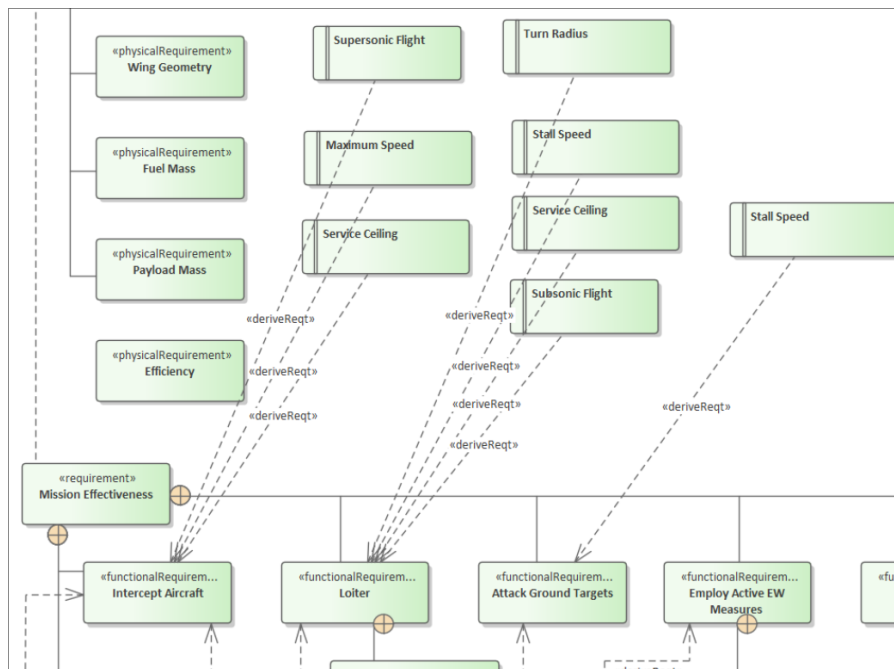


Figure 2: Derivation of Aerial Performance requirements from Mission Effectiveness. Virtualised connector endings have been used for readability.

In addition to functional requirements, some non-functional requirements have been derived. From this paper’s standpoint, a focus has been given to aerodynamic performance requirements, insofar as they enable the highest priority functions (see Section 2.2). values for these performance requirements, where applicable, have been derived from contemporary fighter aircraft data (where it is publically available). Table 1 contains information for five different aircraft [6]: the F-4E Phantom II [7][8][9][10], the F15-C Eagle [11][12][13], the F/A-18 Hornet [11][14][15], the AV-8B Harrier [16][17] (all McDonnell Douglas) and the Grumman A6-A intruder [18]. All are naval aircraft, aligning with the intended market of the UCAV. This data has been used to inform engineering judgement on performance requirements, along with “Physical Characteristics”.

2.2 Requirements Analysis

To determine which requirements were of highest priority, the analytic hierarchy process (AHP) was used. Compared to other potential requirement ranking methods, AHP allows for an individual to determine the priorities from their specific standpoint on a case-by-case basis, which is something that other methods (e.g. \$100) do not allow. Additionally, as the majority of the requirements elicited use the terms “must” or “shall,” it

Table 1: Aircraft Performance and Aerodynamic Parameters

Parameter	F-4E	F-15C	F/A-18A	AV-8B	A-6
p_s	0.356	0.334	0.357	0.219	0.330
$S [m^2]$	50.911	55.649	37.161	21.368	48.310
$C_{L\max}$	1.2	2.1	1.7	1.435	2.1
$m_{pl} [kg]$	1285	1451	3649	2314	907
AR	2.77	3.01	3.52	4	5
$T [N \times No.Engine]$	79620×2	106000×2	71000×2	105868	37000×2
$TSFC \left[\frac{kg}{Ns} \right]$	2.40×10^{-5}	2.07×10^{-5}	N/A	2.24×10^{-5}	2.32×10^{-5}
$m_f [kg]$	5469	6103	5503	3519	3975
C_{D0}	0.0205	0.0218	0.0239	N/A	0.023
$V_{\text{stall}} [ms^{-1}]$	52.4	39.4	60.1	62.5	—
$ROC [ms^{-1}]$	294	390	152	N/A	—
$R [m]$	1825	2193	N/A	807	—
$r_{\text{Turn}} [m]$	280.6	158.5	368.3	397	—

is not possible to differentiate based on language (e.g. MoSCoW). The input relationships can be seen in figure 3. In each comparison, preference was given to requirements aligned with CAP to align with the intended market niche. Autonomy was de-prioritised, as whilst it is an important facet of the UCAV, it has little impact on the airframe, propulsion and outfit. The impacts it does have (e.g. no cockpit, no life support, turns no longer being human limited etc.) have not been considered to manage complexity and should be investigated in a follow up design study.

	Intercept Aircraft	Loiter	Attack Ground Targets	Track Targets	Prioritise Targets	Conduct Beyond Visual Range Combat	Conduct Visual Range Combat	Employ Active EW Countermeasures	Communicate with Operator	Communicate over Datalink	Refuel when Landed	Refuel in Air	Facilitate Maintenance	Land at Airfield	Land on Aircraft Carrier	Land Autonomously	Take Off from Airfield	Take Off from Aircraft Carrier	Take Off Autonomously	Manoeuvre Autonomously
Intercept Aircraft	1	1/3	3	1/3	3	3	1/3	3	3	3	5	3	5	5	7	5	5	7	5	3
Loiter	3	1	5	3	5	5	3	5	5	5	7	5	7	7	9	7	7	9	7	5
Attack Ground Targets	1/3	1/5	1	1/3	1/3	1/3	1/5	1	1	1/3	3	1/3	3	3	5	3	3	5	3	1/3
Track Targets	3	1/3	5	1	3	3	1/3	3	3	3	5	3	5	5	7	5	5	7	5	3
Prioritise Targets	1/3	1/5	3	1/3	1	1	1/5	3	3	3	5	3	5	5	7	5	5	7	5	3
Conduct Beyond Visual Range Combat	1/3	1/5	3	1/3	1	1	1/3	3	3	3	5	3	5	5	7	5	5	7	5	3
Conduct Visual Range Combat	3	1/3	5	3	5	3	1	5	5	5	7	5	7	7	9	7	7	9	7	5
Employ Active EW Countermeasures	1/3	1/5	1	1/3	1/3	1/3	1/5	1	3	3	5	3	5	5	7	5	5	7	5	3
Communicate with Operator	1/3	1/5	1	1/3	1/3	1/3	1/5	1	3	3	5	3	5	5	7	5	5	7	5	3
Communicate over Datalink	1/3	1/5	3	1/3	1/3	1/3	1/5	1	3	3	5	3	5	5	7	5	5	7	5	3
Refuel when Landed	1/5	1/7	1/3	1/5	1/5	1/5	1/7	1/5	1/5	1/5	1	1/3	3	3	5	3	3	5	3	1/3
Refuel in Air	1/3	1/5	3	1/3	1/3	1/3	1/5	1	3	3	5	3	5	7	5	5	7	5	3	3
Facilitate Maintenance	1/5	1/7	1/3	1/5	1/5	1/5	1/7	1/5	1/5	1/5	1/3	1/3	1	3	5	3	3	5	3	1/3
Land at Airfield	1/5	1/7	1/3	1/5	1/5	1/5	1/7	1/5	1/5	1/5	1/3	1/3	1	3	5	3	3	5	3	1/3
Land on Aircraft Carrier	1/7	1/9	1/5	1/7	1/7	1/7	1/9	1/7	1/7	1/7	1/5	1/7	1/5	1/3	1	1/3	1/3	3	1/3	1/5
Land Autonomously	1/5	1/7	1/3	1/5	1/5	1/5	1/7	1/5	1/5	1/5	1/3	1/3	1/3	3	1	3	5	3	1/3	1/3
Take Off from Airfield	1/5	1/7	1/3	1/5	1/5	1/5	1/7	1/5	1/5	1/5	1/3	1/3	1/3	3	1	3	5	3	1/3	1/3
Take Off from Aircraft Carrier	1/7	1/9	1/5	1/7	1/7	1/7	1/9	1/7	1/7	1/7	1/5	1/7	1/5	1/3	1/3	1	1/3	3	1/3	1/5
Take Off Autonomously	1/5	1/7	1/3	1/5	1/5	1/5	1/7	1/5	1/5	1/5	1/3	1/3	1/3	3	1	3	5	3	1/3	1/3
Manoeuvre Autonomously	1/3	1/5	3	1/3	1/3	1/3	1/5	1	3	3	5	3	5	7	9	7	7	9	7	5

Figure 3: Requirement AHP matrix. As values are reciprocal below the diagonal, values above have been emphasised.

A python script was then used to calculate the normalised eigenvector for the respective requirements. These eigenvectors can be seen in table 2. From this, it is clear that the key requirements to focus on are loitering and within visual range combat. This is expected, as they are key components of CAP. This was used as the base for non-functional derivation in Section 2.1, but of note is that the derived requirements often apply to other functional requirements as well (such as Stall Speed relating to Attack Ground Targets). This requirements ranking also indicates where future work can perform more complex design optimisations.

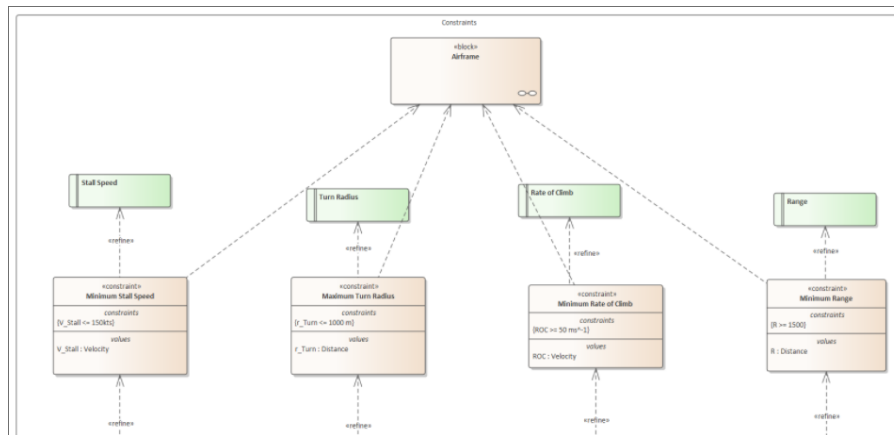
2.3 Design Problems Definition and Specification

From the derived and ranked requirements, a multi-objective optimisation problem was set up such that the optimal combination of the key design variables (DVs) can be found to optimise the design objectives (DOs). The DOs are defined by the requirements; performance requirements set constraints on certain characteristics, and these characteristics are what should be optimised. The DVs are values that the aircraft designer has control over. Linking the two are design enablers (DEs), which are mathematical relationships that provide a value for

Table 2: Ranked requirements and their corresponding weights, to three significant figures.

Rank	Requirement	Weight
1	Loiter	0.169
2	Conduct Visual Range Combat	0.146
3	Track Targets	0.0980
4	Intercept Aircraft	0.0852
5	Conduct Beyond Visual Range Combat	0.0698
6	Prioritise Targets	0.0690
7	Employ Active EW Countermeasures	0.0574
8	Communicate with Operator	0.0513
9	Communicate over Datalink	0.0482
10	Refuel in Air	0.0400
11	Manoeuvre Autonomously	0.0304
12	Attack Ground Targets	0.0295
13	Refuel when Landed	0.0208
14	Facilitate Maintenance	0.0186
15	Land at Airfield	0.0158
16	Land Autonomously	0.0141
17	Take Off from Airfield	0.0126
18	Take Off Autonomously	0.0108
19	Land on Aircraft Carrier	0.00751
20	Take Off from Aircraft Carrier	0.00649

a DO from a combination of DVs. The combination of DVs, DEs, and DOs is termed a design problem. The key performance requirements are stall speed (V_{stall}), turn radius (r_{turn}), range (R), and rate of climb (ROC). These have been modelled using «constraint» blocks in SysML, as is seen within the Constraints boundary on the Requirements diagram (see figure 4).

**Figure 4:** DO constraint relationships to requirements and architecture

These constraints are then refined by constraint blocks containing the DEs. The DEs are based on classical aerodynamic equations. Stall speed is derived from the well-known lift equation [19]:

$$L = \frac{1}{2} \rho V^2 S C_L$$

where L is lift, ρ is the density of air, V the freestream speed, S the wetted wing area, and C_L the lift coefficient. In steady flight, the aircraft is producing lift equal to its weight, such that it does not sink. At stall, V

is set as low as it can be. Therefore, as ρ and S do not change, C_L must be maximised, and this limit is what defines the stall speed.

$$W = \frac{1}{2}\rho V_{stall}^2 S C_{Lmax}$$

$$\Rightarrow V_{stall} = \left(\frac{2W}{\rho S C_{Lmax}} \right)^{\frac{1}{2}}$$

To determine turn radius a similar condition is used [20]. With the maximum wing loading $n_{max} \equiv \frac{L_{max}}{W}$,

$$r_{turn} = \frac{V^2}{g(n_{max}^2 - 1)^{\frac{1}{2}}}$$

Determining rate of climb also considers the thrust available to the aircraft. Specifically, it requires the excess thrust - the amount of thrust that exceeds the drag at a given freestream speed. It is calculated using the following equation [21]:

$$ROC = \frac{(T - D)V}{W}$$

where T is the total thrust, D the calculated drag, and W the weight. Drag can be calculated in the same manner as lift:

$$D = \frac{1}{2}\rho V^2 S C_D$$

C_{Lmax} is a function of aircraft geometry, whereas C_D is actually composed of multiple parts, however: the zero lift drag, and the lift-induced drag. This can be represented with the *drag polar*:

$$C_D = C_{D0} + \frac{C_L^2}{\pi e AR}$$

where C_{D0} is the zero lift drag coefficient (another geometric property), AR the aspect ratio of the aircraft (a measure of a wing's "slenderness"), and e the Oswald efficiency factor. e is a measure of how closely a wing's lift distribution is to a purely elliptical distribution; for the purpose of this paper, a constant value of 0.8 is reasonable [22]. Now, whilst C_{Lmax} is a geometric property, C_L is a function of the freestream velocity as the aircraft will only produce as much lift as is needed to counteract its weight.

$$C_L = \frac{2W}{\rho S V^2}$$

Finally, the range is calculated using the Breguet Range Equation [23]:

$$R = \frac{V}{TSFC} \frac{C_L}{C_D} \ln \left(\frac{W_i}{W_f} \right)$$

Where W_i is the initial weight of the aircraft at take off (i.e. fuel, payload, and structure) and W_f is the final weight (i.e. payload and structure, with all fuel burnt). For this paper, all instances of W have been W_i , and that can be calculated from the mass of the fuel m_f , mass of payload m_{pl} , acceleration due to gravity $g \equiv 9.81$ and proportion of the aircraft's overall weight which is structure p_s (from [6]):

$$W = \frac{g(m_f + m_{pl})}{1 - p_s}$$

TSFC is the thrust specific fuel consumption, and is a measure of the fuel used to produce a unit thrust. It is normally published by the engine manufacturer or identified in reports (see table 1)

To represent these in SysML, constraint blocks were again used with the equations as constraints, and the DVs as values. To do this, however, the DVs must exist somewhere in the model. A simple physical architecture was generated in the "Architecture" diagram. This architecture is composed of four subsystems, with the key subsystem being the airframe (as all of the DOs are in some way aerodynamic metrics). These subsystems are decomposed into lower level parts, which at this point are present for representation. The airframe subsystem then contains value properties, which correspond to the performance metrics (DOs) but also the DVs. Other

values related to other subsystems (e.g. m_f) were assigned to their respective blocks, and ports permit the transfer of this information to the airframe block. This is so that they can appear in parametric diagrams, which are owned by the airframe block. Each DV is given bounds with constraint blocks, based on the data in table 1. Each DO has a corresponding parametric diagram, which shows diagrammatically the flow of DVs through to the constraints set by the performance requirements, enabling traceability throughout the model. The diagrams are titled Range, Rate of Climb, Stall Speed, and Turn Radius. An example is shown in figure 5

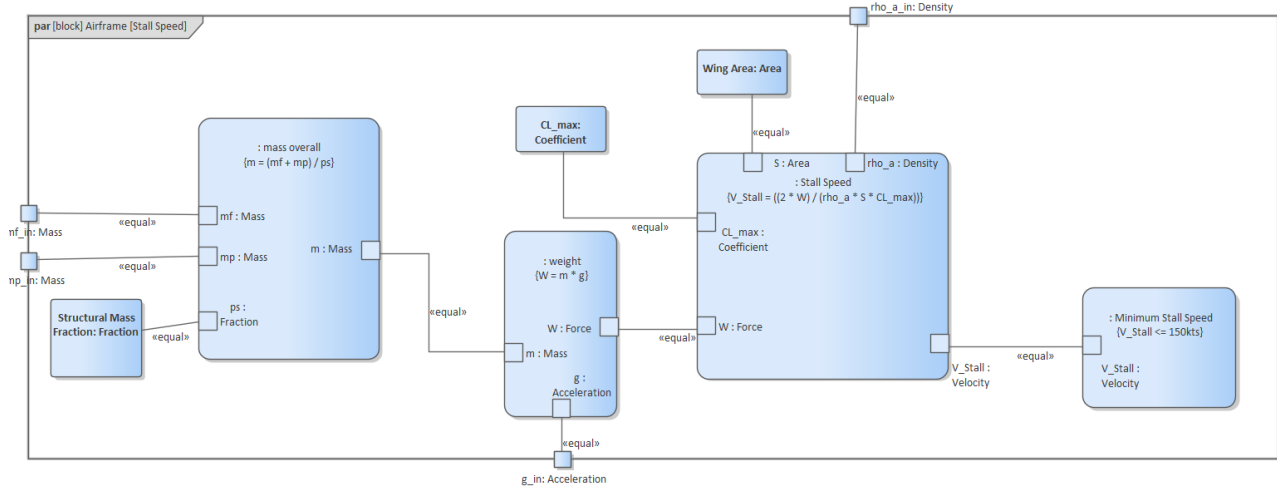


Figure 5: Stall Speed Parametric Diagram

2.4 Design Analysis Plan and Methodology

For analysis and visualisation, these DEs were coded in the Python programming language. Python has many modules which allow for scientific computation and visualisation, namely numpy, pandas, and matplotlib. These tools were used to display a matrix of scatter plots, showing pairwise relation between DVs and DVs, DVs and DOs, and DOs and DOs. To generate the points, the module pydoe was used to implement Latin Hypercube Sampling, taking a number of points between the bounds and finding the corresponding DO values. Additionally, scikit-learn allows for machine learning algorithms to be used. In this case, it was used to perform ordinary least squares regression to provide partial dependency plots, which are plots essentially showing the sensitivity of the DOs to each DV, mapping a response surface.

Figures 6 and 7 show this visualisation.

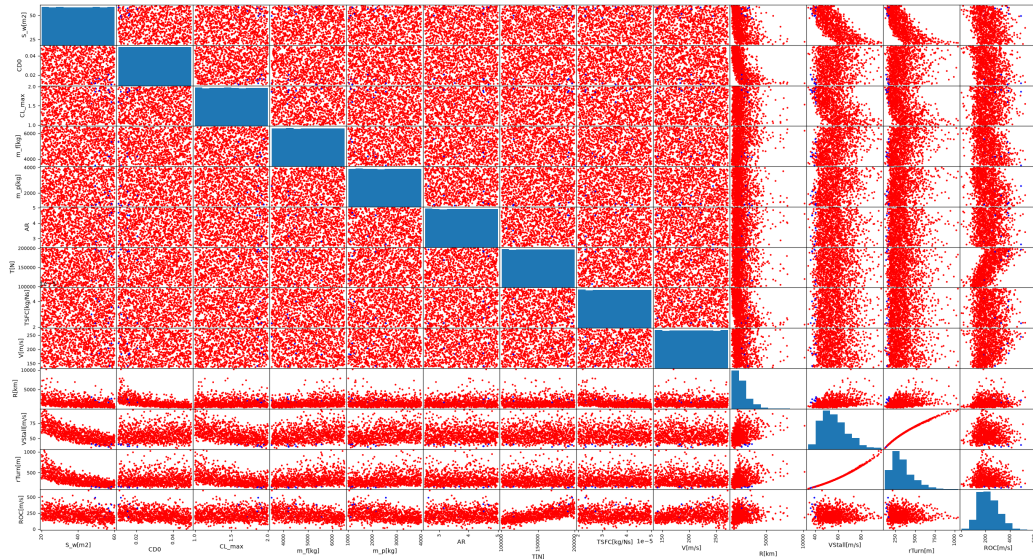


Figure 6: Initial design space visualisation

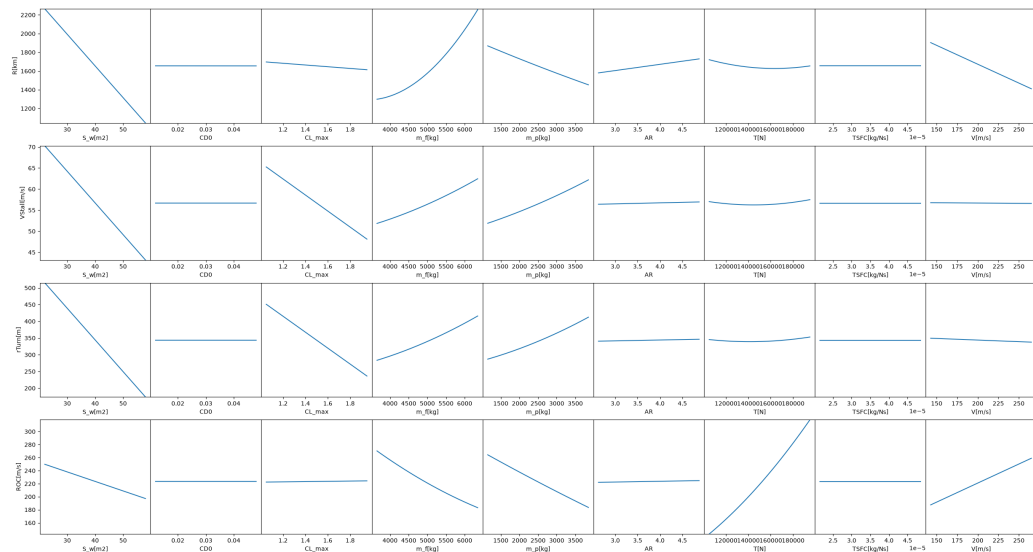


Figure 7: Sensitivity analysis of the design space

Attempting to optimise all 9 DVs simultaneously would be computationally expensive and potentially impossible. Instead, fixing certain DVs and optimising the remaining ones individually is simpler and more likely to produce a suitable result. In the axiomatic design methodology, this represents the mapping from functional space to physical space. The DOs were ranked based on the number of dependencies they had (with the fewest at the top), and the DVs with the most interactions were shifted to the left. Any cases where the sensitivities were negligible were made blank. This resulted in the matrix shown in figure 8.

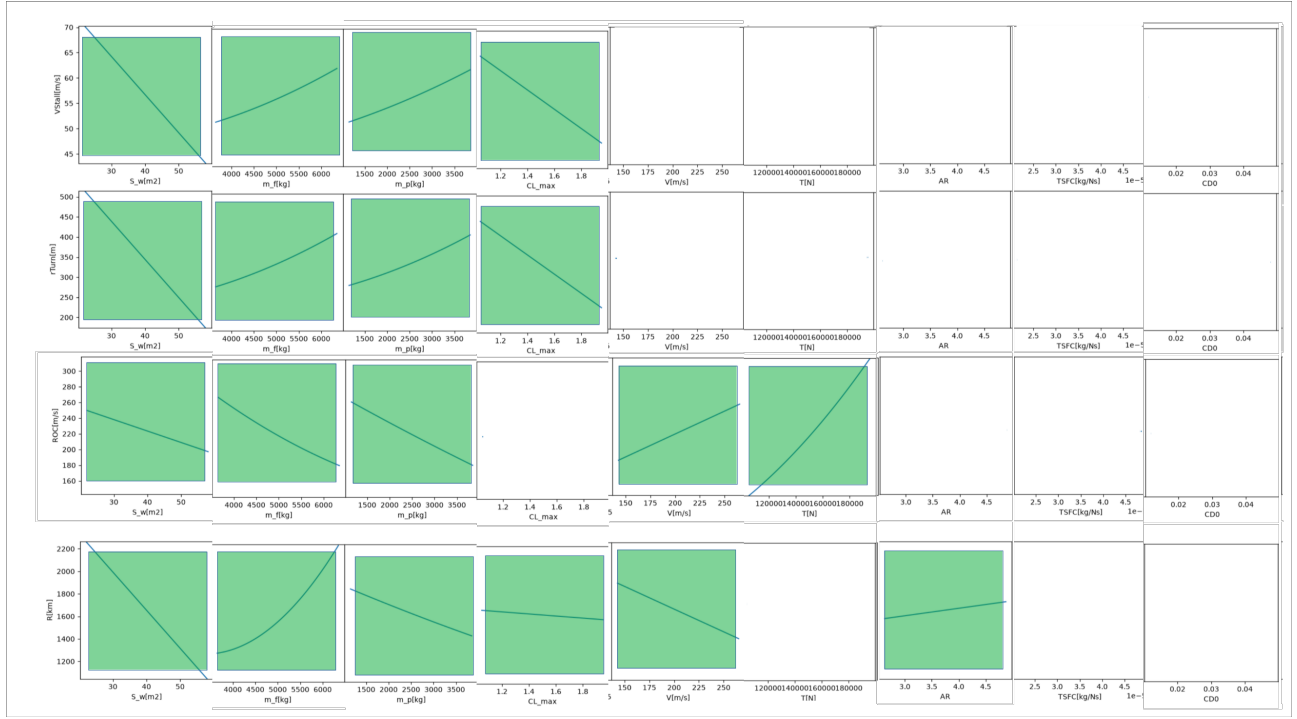


Figure 8: Sensitivity matrix post-ranking

There are a few conclusions that can be drawn from this. Firstly, TSFC and C_{D0} have negligible impact on all of the DOs. This implies that they are dominated by the other DVs in each DE. It is possible that this is simply a feature of the response surface, with different bounds of the DVs having different sensitivities, but as the bounds are based on contemporary aircraft these DVs shall be discounted and fixed at their median value. Secondly, AR and T only meaningfully affect one DO each. This is important, as it means that they can be optimised by simply fixing the DV at the appropriate boundary: in this case, Thrust is set to $200kN$ and AR set to 5. Thirdly, increasing m_p has a detrimental effect on all four DOs. This is similar to AR and T , in that it can be optimised by fixing it at the lowest bound ($1000kg$).

This leaves four design variables with no clear option to fix. In a decoupled system, there would always be a clear diagonal in a square matrix, making it easy to determine which DV should next be fixed. Instead, as there is no diagonal and too many DVs to make a square matrix, a different approach must be taken. In figure 6, there are a few passing combinations of DVs (in blue). This means that the orthotope of feasible designs exists within the current bounds. To constrain the design space, the bounds for S were set to the most extreme values that still passed, which were $31 \leq S \leq 58 [m^2]$. From this point, each design variable's bounds were to be constrained to the extremes of the passing combinations iteratively until an entirely passing orthotope was seen.

2.5 Design Trade-offs Visualisation

As mentioned, the bounds of S were reduced to $31 \leq S \leq 58 [m^2]$. That resulted in figure 9

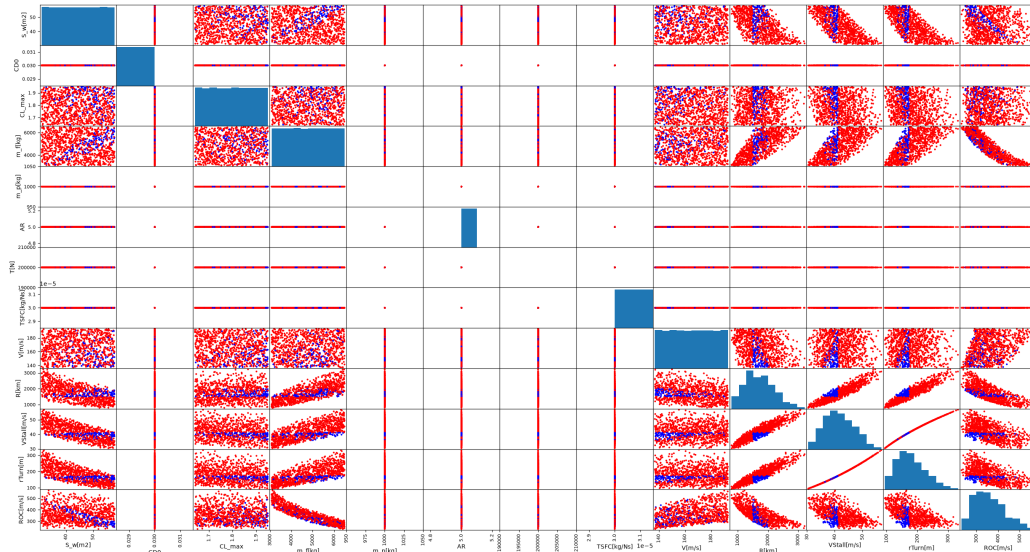


Figure 9: First round design space visualisation with reduced S

In the S against m_f plot, a diagonal band of passing values can be seen extending from the bottom left corner to the top right. In the other plots, clouds of passes exist in the same space as failures. As such, no simple bounds can be seen of a feasible orthotope. The bounds of m_f and S were then reduced simultaneously, to focus in on a rectangle within the diagonal band, resulting in figure 10

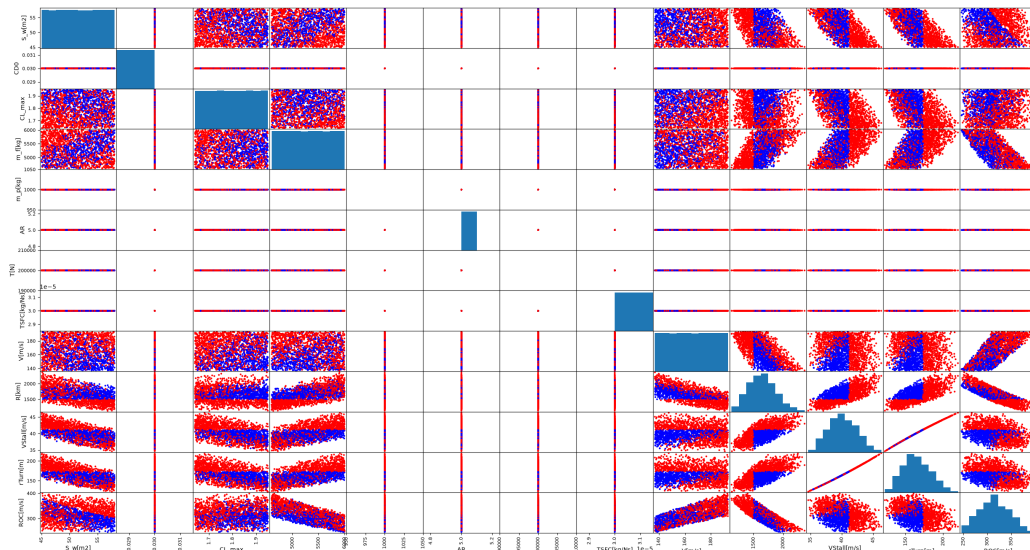


Figure 10: Second round design space visualisation with reduced S , m_f

This process was repeated 6 times, with figures 11-14 showing the stages of visualisation.

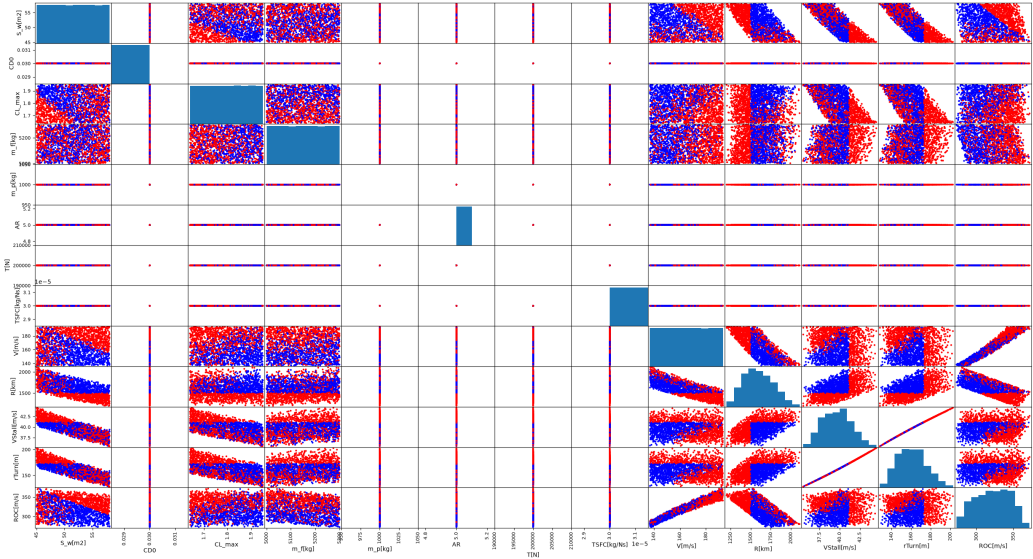


Figure 11: Third round design space visualisation with reduced S, m_f

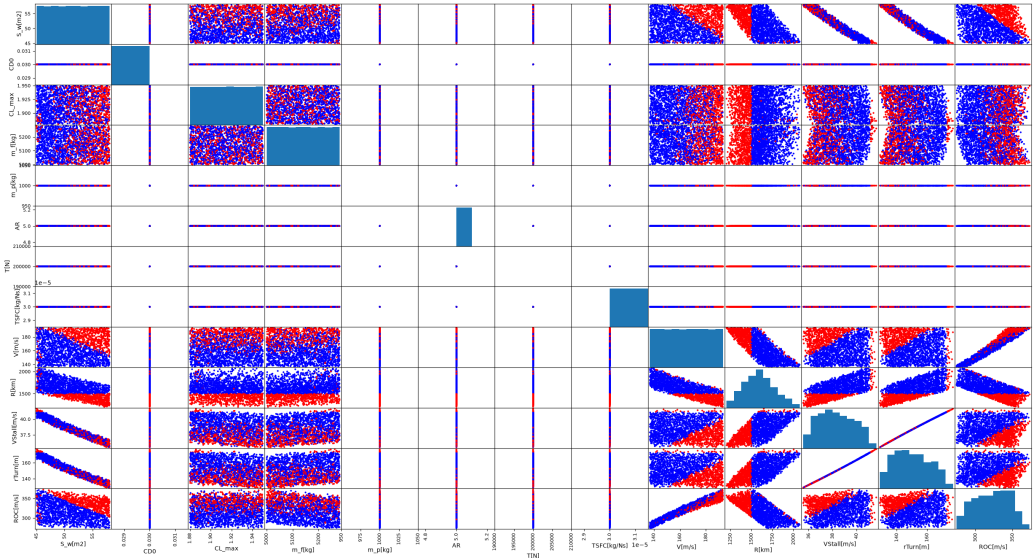


Figure 12: Fourth round design space visualisation with reduced C_{Lmax}

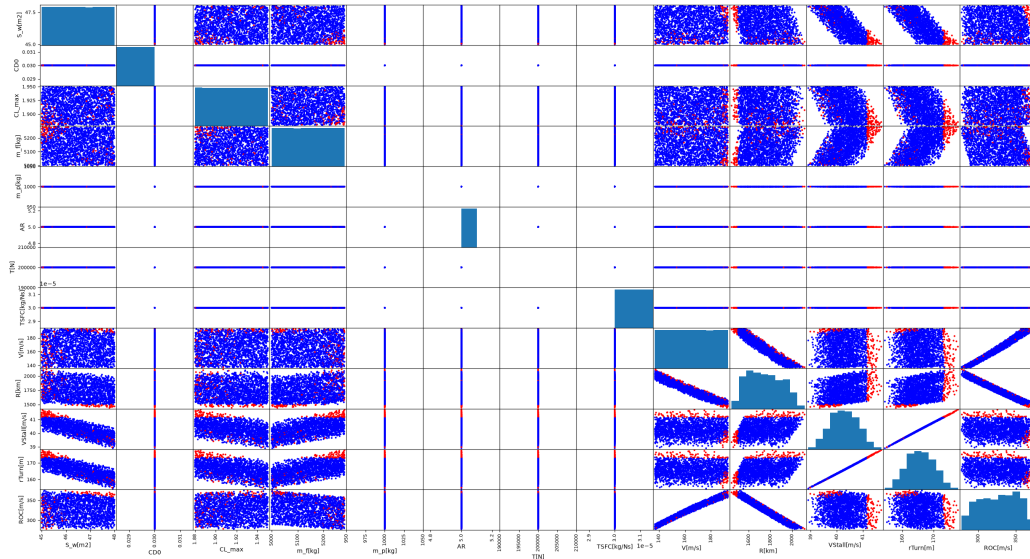


Figure 13: Fifth round design space visualisation with reduced S

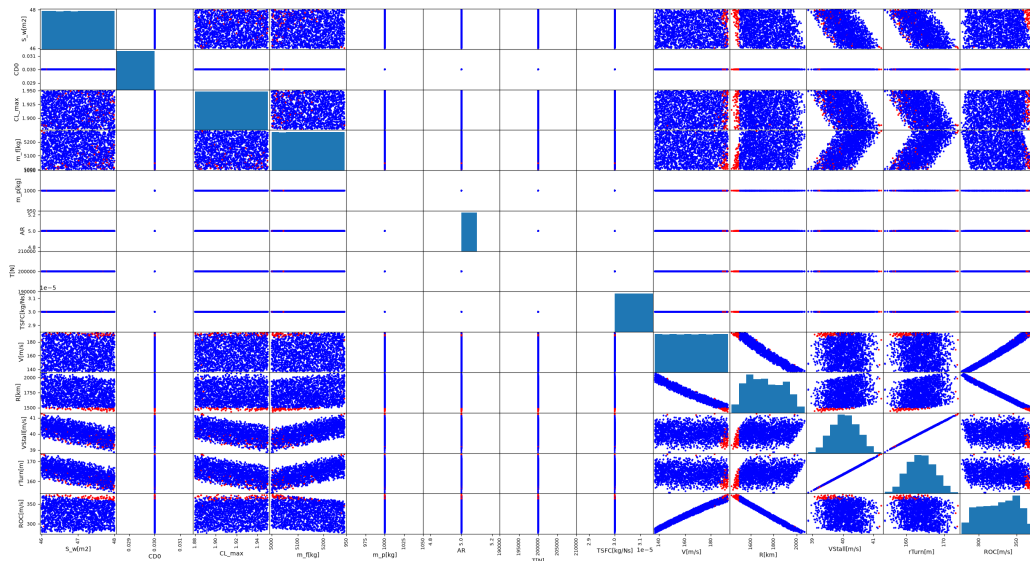


Figure 14: Sixth round design space visualisation with reduced S

At this point, a cluster of failures can be seen in the corners of the DV-DV plots. They are intermixed with the passes, so there is no clear way to move the bounds to exclude them. This implies that there is a part of the 4D orthotope that is extending into the failure region, and that it is being displayed unusually in 2D space. Instead of attempting to constrain the orthotope, it was decided that V be constrained to exclude the failing points, as it was the design variable with the greatest range between bounds remaining. Additionally, the UCAV must perform across the flight envelope - constraining V clearly implies that the design is non compliant. Instead, future work would argue that the performance requirements are too stringent for the entire flight envelope, and negotiate a compromise where the requirements are relaxed either across the envelope, or only for the regions where the designs are infeasible. This constraint results in figure 15 with no failures.

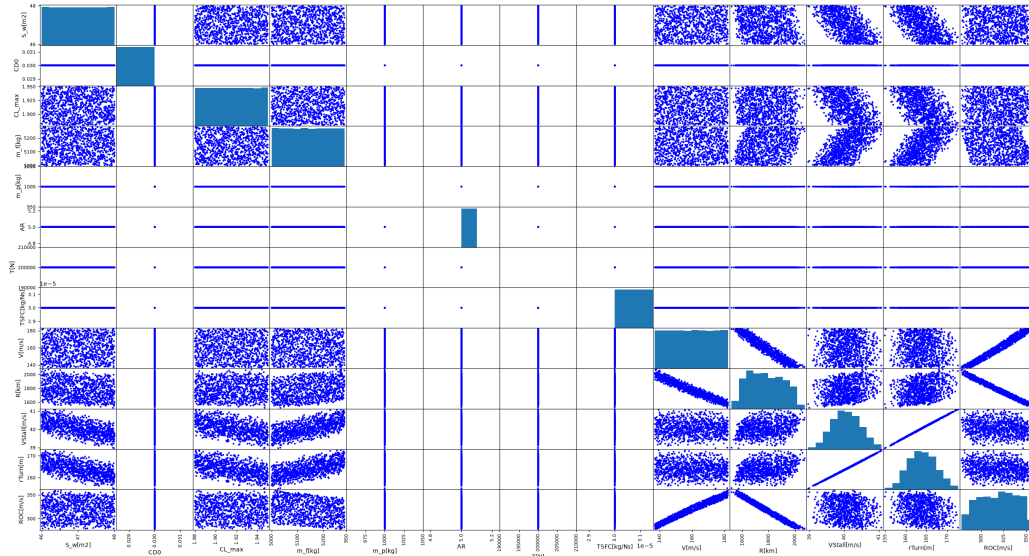


Figure 15: Final design space visualisation with final bounds set

2.6 Design Trade-offs Analysis

Now that a feasibility region has been defined, it is possible to determine the final solution specification. There are two methods of achieving this, that are determined by whether or not there is preference in DVs or in DOs. In this early stage, there is no clear preference in DVs. All of variables are free to move within their bounds, and there are no requirements forcing any to be minimised/maximised. This is a feature of the restricted requirements set and the requirement rankings, as introducing cost, efficiency, or weapons loadout would almost certainly have implications for the DVs. As it stands, though, no DV is preferred over another, but there is preference in the DOs. It would be possible to determine which performance requirements are derived from the most capabilities, but as the CAP capability is so much more preferred, selecting the most important DO to enable that is more suitable. On the assumption that the majority of the time a combat air patrol will not encounter hostile aircraft, the most important objective is range; range is considered on every sortie, not only those where combat occurs.

As such, a heuristic search for optimal values was performed. This was done with a multi-objective optimisation genetic algorithm, implemented by the pymoo module. Pymoo has many available algorithms, but for this optimisation NSGA-II was chosen. This was implemented in the `/new-bounds/optimizeMultiple-orthotope.py` script, which performs 2000 iterations on a population size of 200. This produced a Pareto-optimal front of DVs, which was then coloured by the resultant Range. See figure 16. This script also output the design with the greatest range.

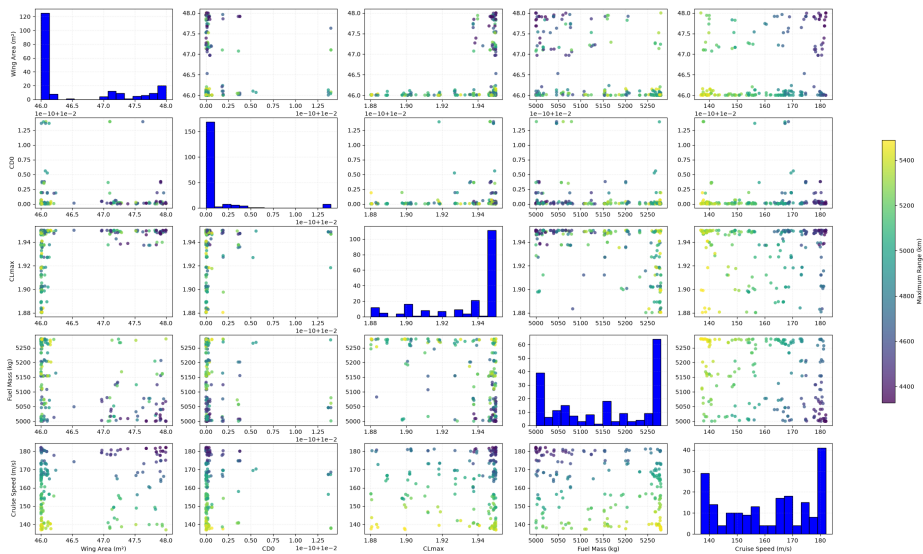


Figure 16: Pareto-optimal design variables, with range visible

Should some DVs become preferred over others, a constraint based analysis would be more suitable, showing robust regions instead of a single value.

3 Conclusion

3.1 Design Specification

Following the trade-off and optimisation analysis, the final system has the following values for implementation:

Table 3: Aircraft parameters

Design Variable	Value
$S [m^2]$	46
C_{D0}	0.01
C_{Lmax}	1.95
$m_f [kg]$	5280
$m_p [kg]$	1000
AR	5
$T [N]$	200,000
$TSFC \left[\frac{kg}{Ns} \right]$	3.00×10^{-5}
$V [ms^{-1}]$	137

Which produces:

Table 4: Aircraft performance parameters

Design Objective	Value
$R [km]$	5288
$V_{Stall} [ms^{-1}]$	41
$r_{Turn} [m]$	169
$ROC [ms^{-1}]$	293

3.2 Achievement of Aims and Objectives

This paper achieves its objectives by:

1. Providing a feasible architecture in SysML along with requirement modelling and parametric diagrams.
2. Reducing wide bounds on DVs to a set of optimal values for a given DO
3. Being traceable with relationships throughout the model

As such, this paper achieves its objective, and provides a set of design variables that produce a UCAV that meets the problem statement and narrative.

3.3 Highlights and Recommendations

Of particular note in this work is the development of a capabilities model, which can be traced through entirely to the final specification. This makes the decisions made throughout easier to defend in a design review. Additionally, this work arms a design engineer or team in negotiations with the customer. As it is occurring early in the project lifecycle, it is still at a point where issues and sticking points can become new requirements instead of non-compliances, potentially having contractual and financial implications.

First among the recommendations is the need to fully understand the system from multiple perspectives. At this time, the design optimisation has been performed with a significant aerodynamic bias, which would be mitigated with interdisciplinary experts collaborating on modelling, requirements ranking, and optimisation. A second recommendation is that this work is furthered and iterated on throughout the project. Having more

fidelity in the model has obvious benefits for design work, but also performing this consistently when changes occur to requirements and design variable bounds prevents outdated and incorrect assumptions being held from early results affecting the design as it progresses.

3.4 Bibliography

It is recommended that the reader consults *Aircraft Design: a conceptual approach* by Daniel P. Raymer, and also that they consult the other volumes of Jan Roskam's *Aircraft Design* series.

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